
CSE 457

Machine Learning

Ensemble Methods

Fall 2025

How to select a movie?

Suppose you want to watch a movie, how can you decide?

- You can ask several friends and come to a decision.
- Its better to ask more than one.
- strong law of large numbers

IDEA: Combine single classifier results!

Wisdom of crowd

Approaches

- Voting
 - Majority
 - Average
 - Weighted
- Bagging
- Stacking
- Cascading
- **Boosting**

Basic Assumption

Base Classifiers - single classifiers that are used

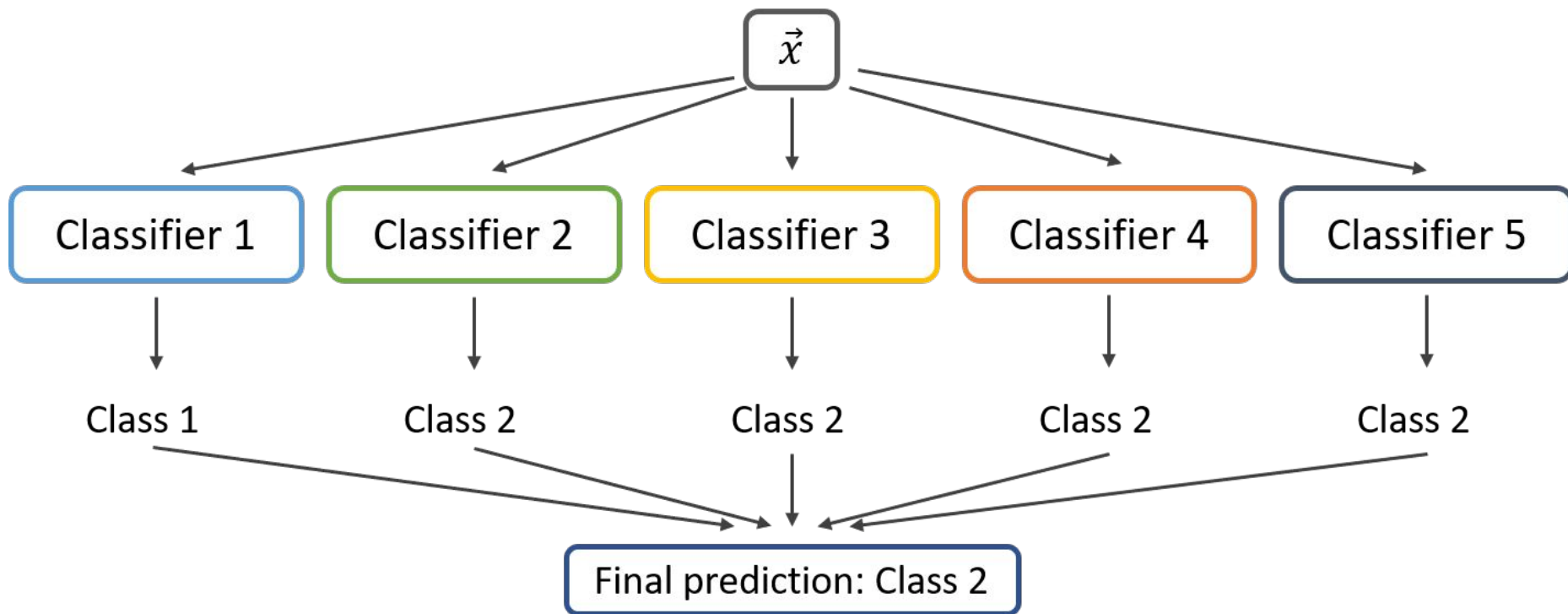
Ensemble - a collection of base classifiers

Either

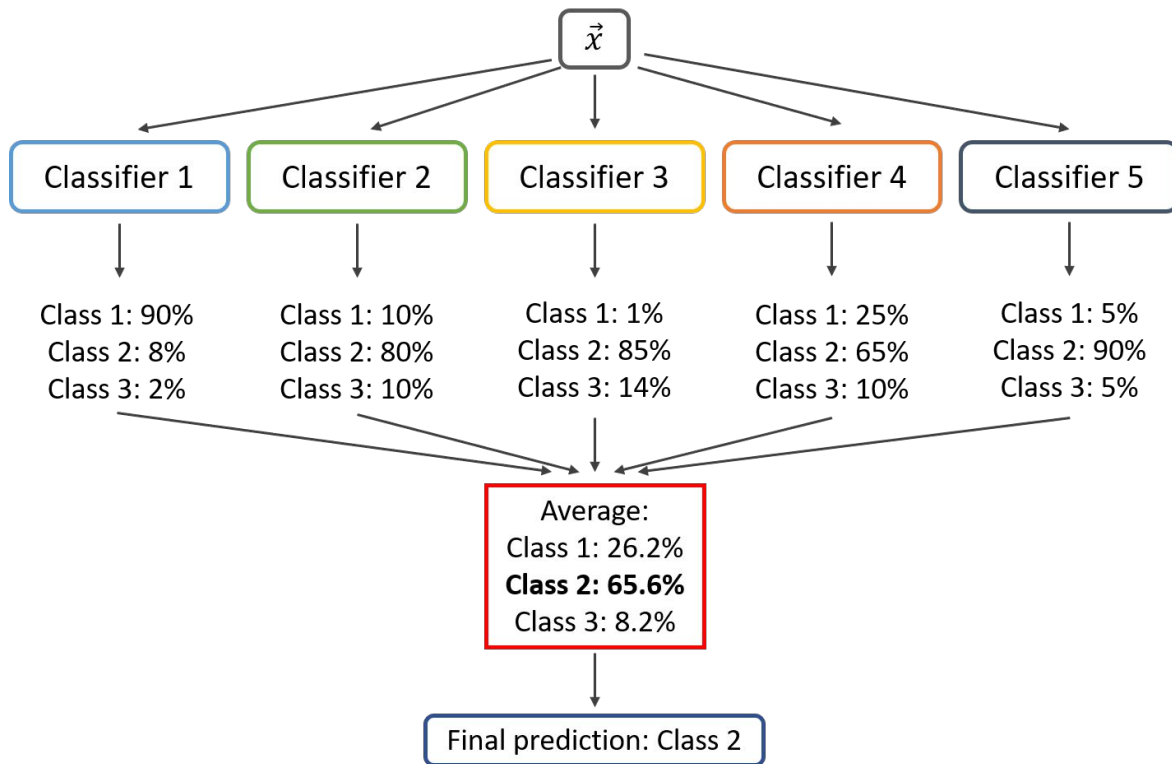
- Models should be different
- Data should be different

[remember ensemble of naive bayes does not work]

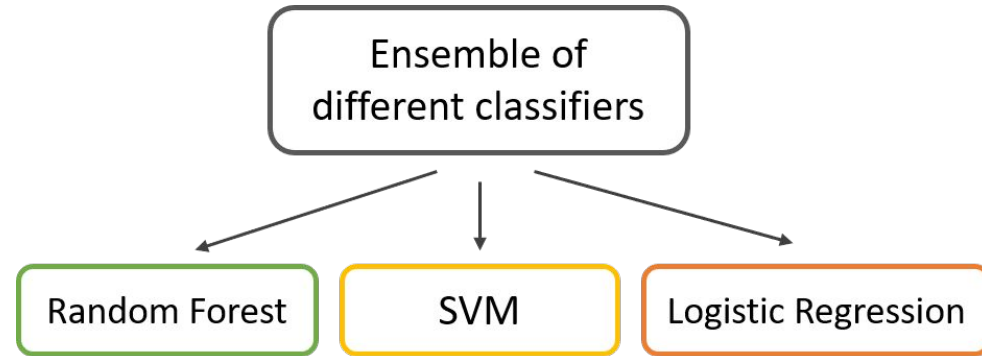
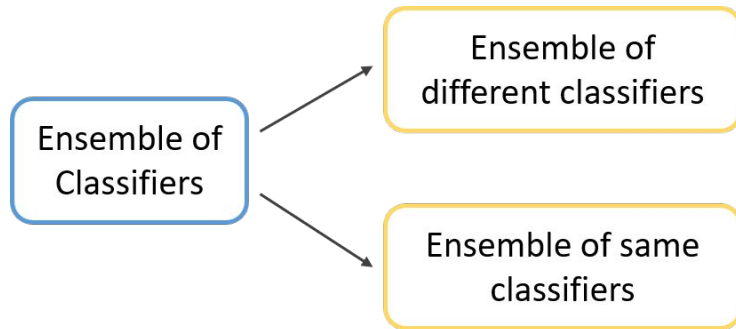
Majority Voting



Hard Voting vs Soft Voting



Base Classifiers in Ensemble



Bootstrap Sampling

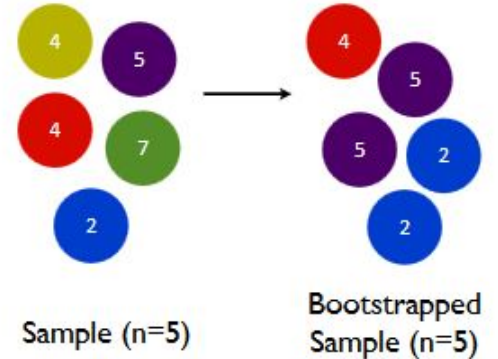
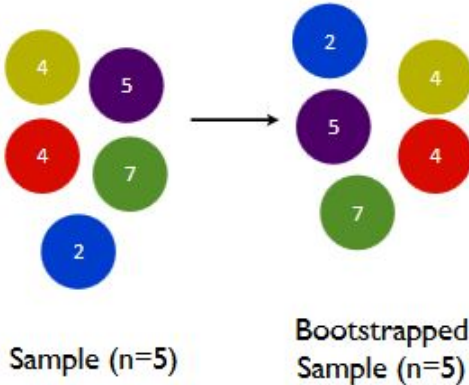
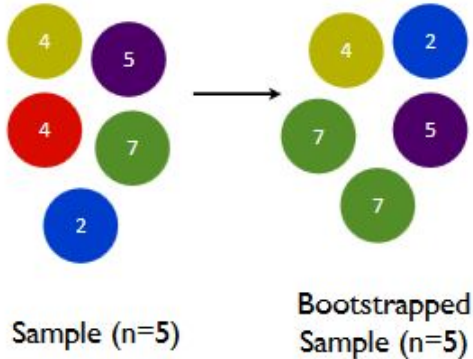
$$1, 2, 3, 4, 100 = 22$$

$$1, 2, 2, 3, 4 = 2....$$

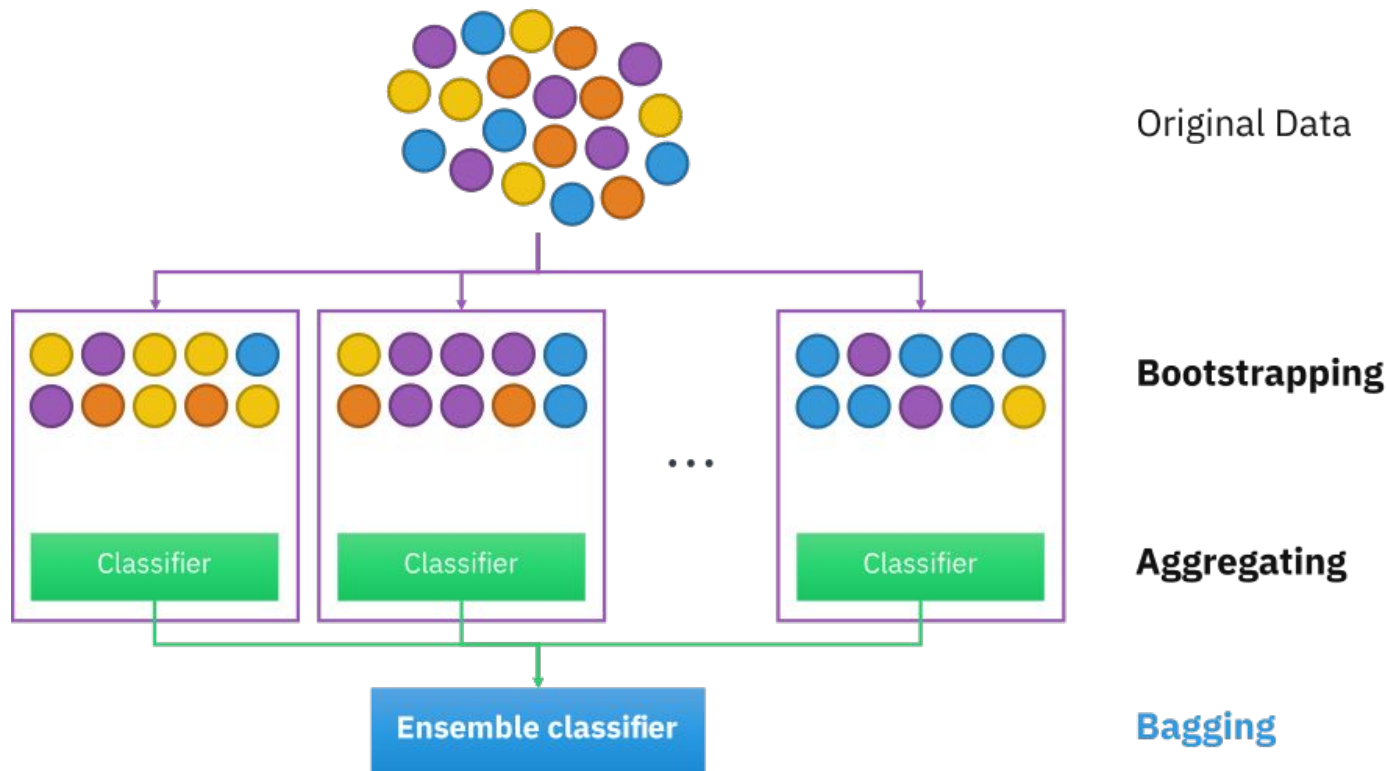
$$1, 1, 3, 4, 100 = 21...$$

$$1, 2, 3, 3, 100 = 21. ...$$

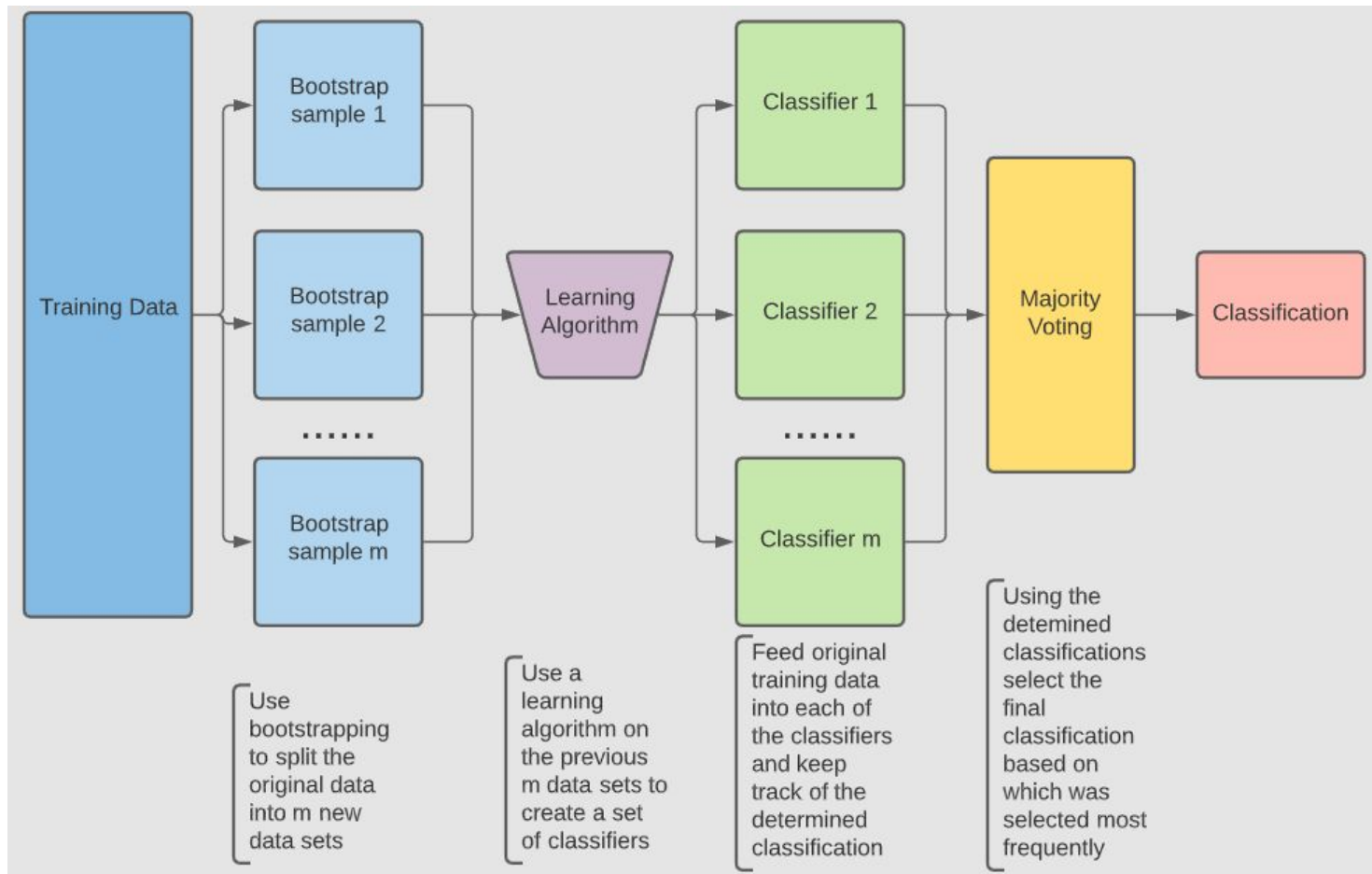
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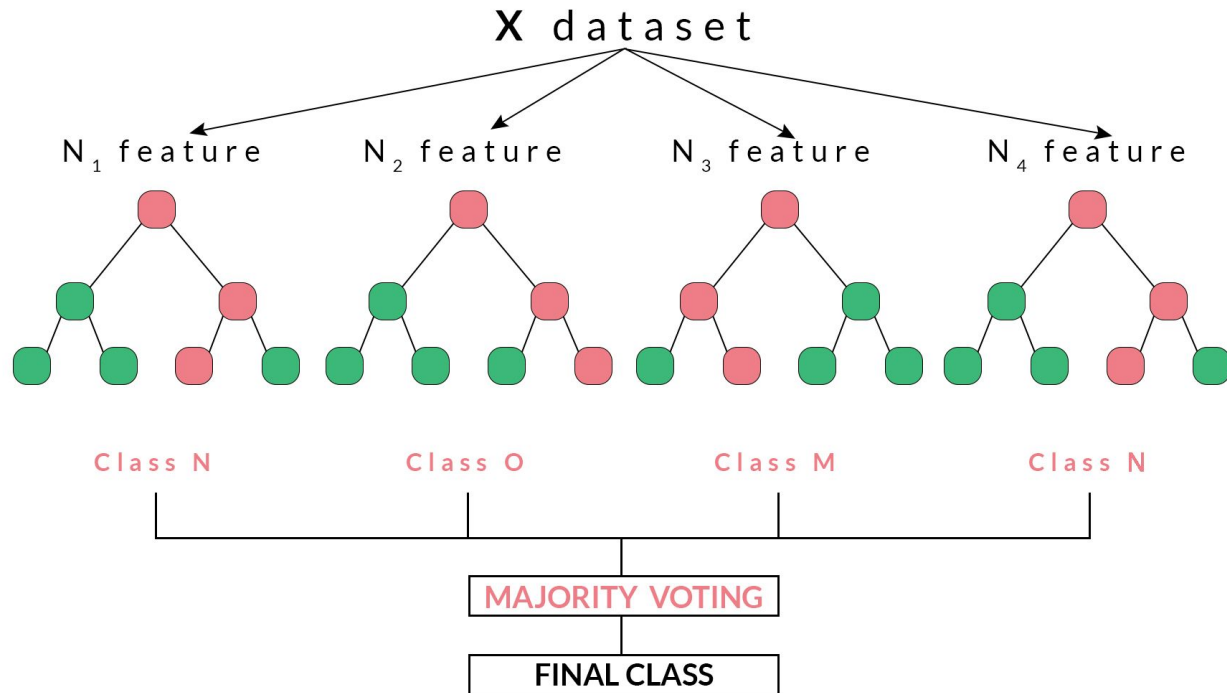
Bagging == Bootstrap Aggregating



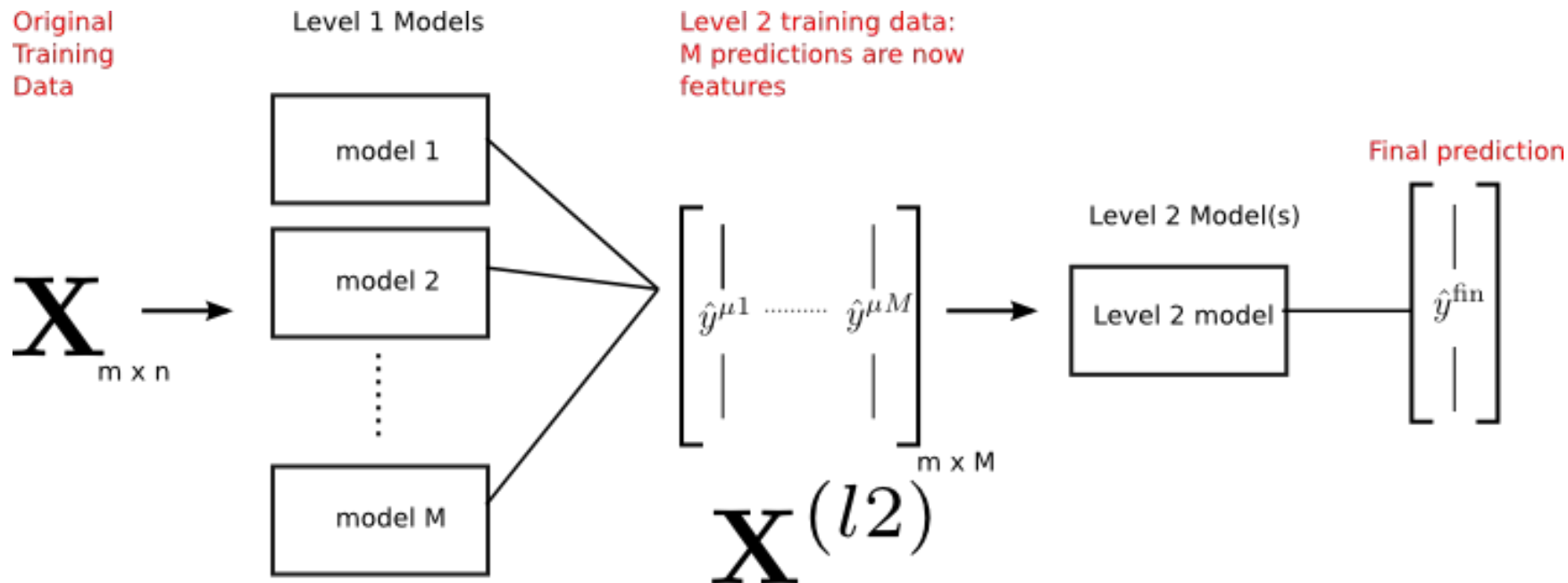
BAGGING



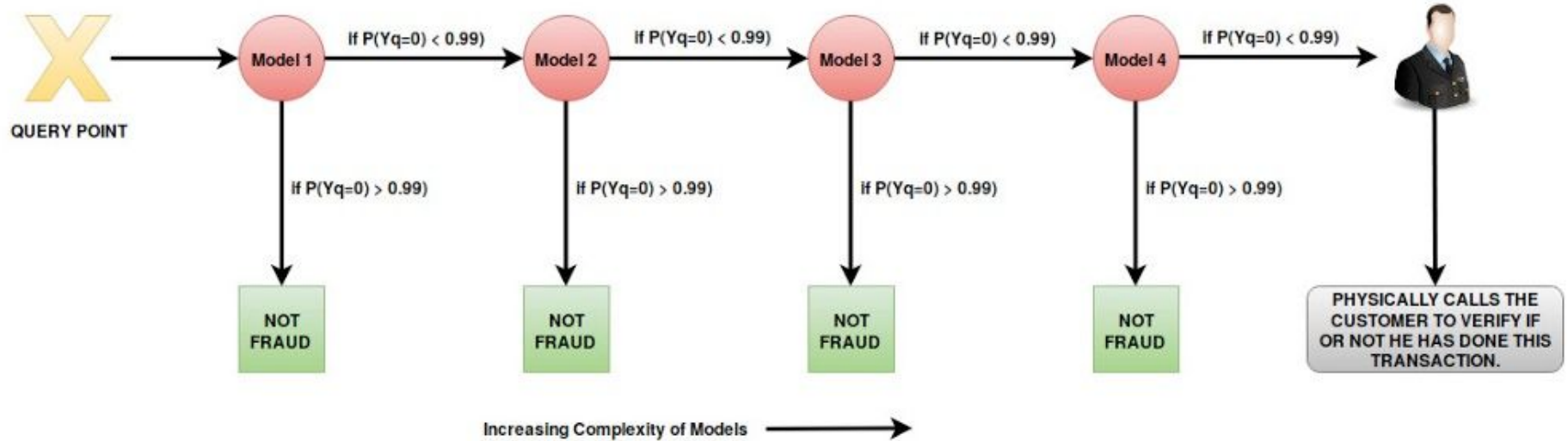
Random Forest



Stacking



Cascading



DIFFERENT STAGES OF QUERING CASCADE CLASSIFIERS IN A FOUR MODEL CASCADE SYSTEM

Boosting

Adaptive Boosting == AdaBoost

Robert Elias Schapire is an American [computer scientist](#), former David M. Siegel '83 Professor in the [computer science](#) department at [Princeton University](#), and has recently moved to [Microsoft Research](#). His primary specialty is theoretical and applied [machine learning](#).

His work led to the development of the [boosting ensemble algorithm](#) used in machine learning. Together with [Yoav Freund](#), he invented the [AdaBoost](#) algorithm in 1996. They both received the [Gödel prize](#) in 2003 for this work.



A single classifier

A single classifier produces a label
[-1,+1]

$h(x)$

Error = e = percentage of mistakenly
classified samples [0,1]

1 == all wrong

0 == all correct

0.5 == random classifier



Think of a balanced case, a classifier that
produces single label for all, what is the
error?

What is the error for a random classifier?

A weak classifier

If error < 0.5 it is a weak classifier

Strong classifier

classifiers with even more good accuracies and less errors



Think of a balanced case, a classifier that produces single label for all, what is the error?

What is the error for a random classifier?

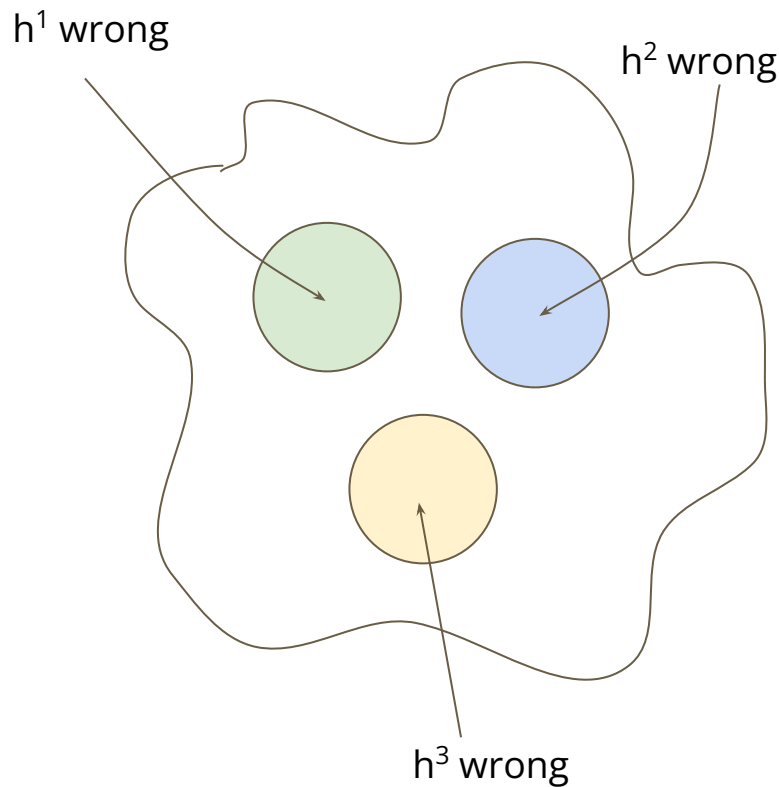
Ensemble

Ensemble

$$H(x) = \text{sign} [h^1(x) + h^2(x) + h^3(x)]$$

Now in any region, where one is wrong, the other two provides correct answer.

Thus we can build a perfect ensemble!



Is it always like that? NO

Ensemble

Ensemble

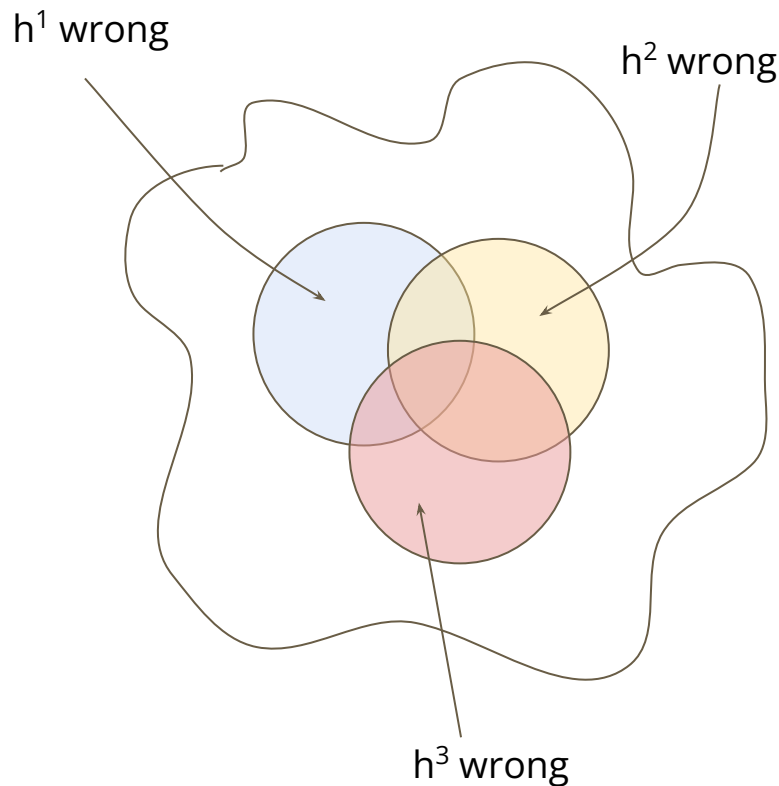
$$H(x) = \text{sign} [h^1(x) + h^2(x) + h^3(x)]$$

In the overlapping areas, the ensemble performs worse than the individuals!

(what if this region is large?)

(what if this region can be minimized?)

(what if we can change their importance?)



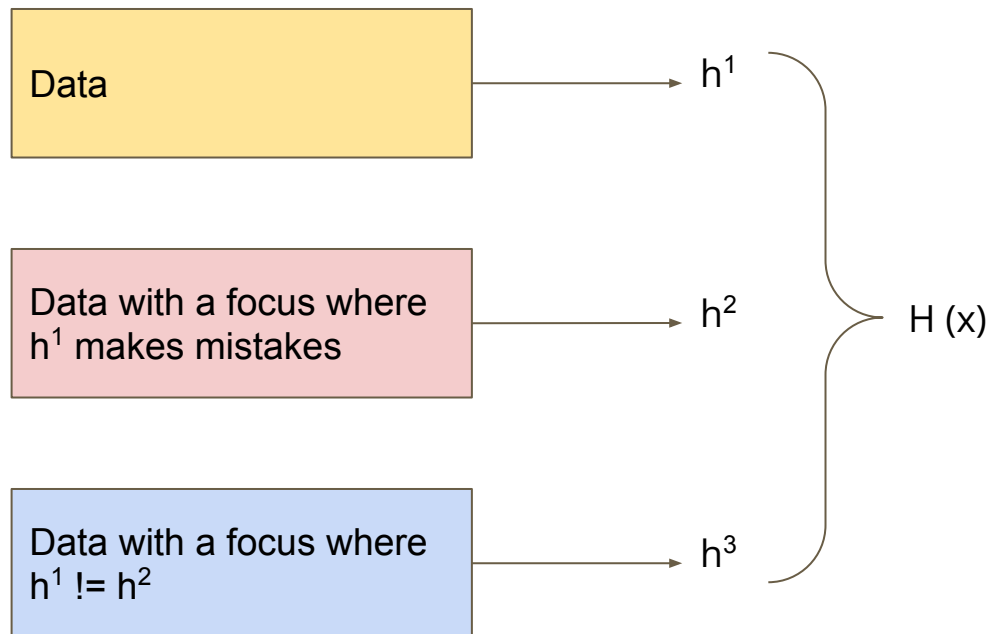
Is it always like that? NO

Ensemble

- Change data?
- Change Voting or ensemble organization?

Focus == weights

Initially all equal



Error Calculations

Suppose there are N items

Now, initially all instances are of similar importance

Lets assume they have weights w_i

We can write error as sum of weights of all instances wrongly classified.

$$e = \frac{1}{N} \sum_{wrong} .1 = \sum_{wrong} \frac{1}{N}$$

$$w_i = \frac{1}{N}$$

$$e = \sum_{wrong} w_i$$

Example

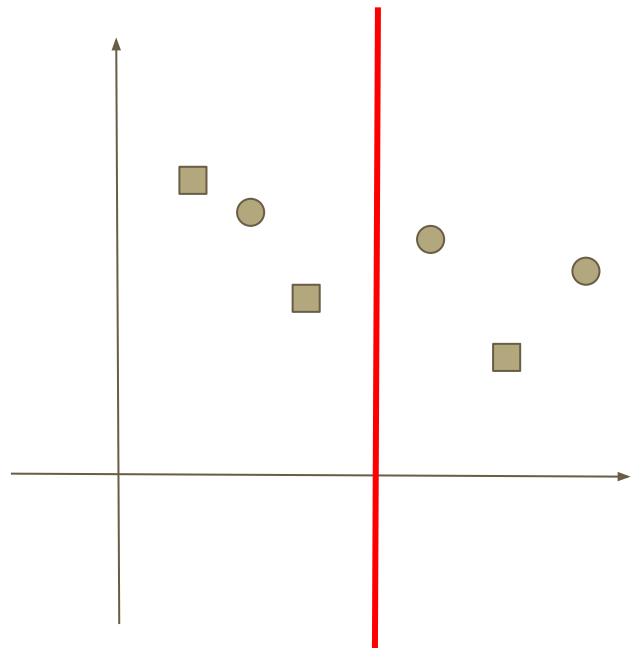
Two instances are wrongly classified

Error = ?

Total weight = ?

Accuracy = ?

What happens if the weights are different?



distribution



$$\sum_{all} w_i = 1$$

We will always try to keep it as a distribution

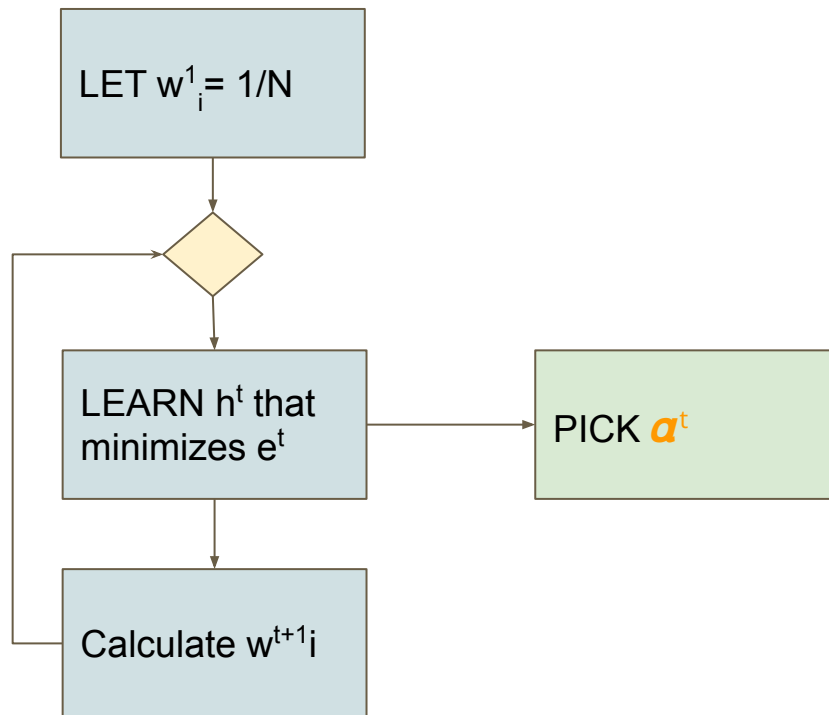
Iteratively build the ensemble

Ensemble

$$H(x) = \text{sign} [\alpha^1 h^1(x) + \alpha^2 h^2(x) + \alpha^3 h^3(x) + \alpha^4 h^4(x) + \dots]$$

So lets start building the algorithm from beginning...

Iterative Algorithm: Boosting



two issues

- 1) How to pick α ?
- 2) How to calculate w in the next iteration?

Updating Weights

$$w_i^{t+1} = \frac{w_i^t}{Z} e^{-\alpha^t \underbrace{h^t(x)}_{\text{prediction}} \underbrace{y(x)}_{\text{label}}}$$

some normalizing factor

w increases? When?

think of h and y

w decreases? When?

think of h and y

**Z is required since
multiplying by
exponential will
not guarantee the
distribution**



Min Error Bound

$$\alpha^t = \frac{1}{2} \ln \frac{1 - e^t}{e^t}$$

Lets get some intuitions
on this expression



That works!

Weight Updates

$$w_i^{t+1} = \frac{w_i^t}{Z} e^{-\alpha^t h^t(x) y(x)}$$

$$\alpha^t = \frac{1}{2} \ln \frac{1 - e^t}{e^t}$$

$$w_i^{t+1} = \frac{w_i^t}{Z} \times \begin{cases} \sqrt{\frac{e^t}{1 - e^t}} & \text{correct} \\ \sqrt{\frac{1 - e^t}{e^t}} & \text{wrong} \end{cases}$$

Without Z

$$w_i^{t+1} = w_i^t \times \begin{cases} \sqrt{\frac{e^t}{1-e^t}} & \text{correct} \\ \sqrt{\frac{1-e^t}{e^t}} & \text{wrong} \end{cases}$$

$$\begin{aligned} \sum_{\text{correct}} w_i^{t+1} &= \sqrt{\frac{e^t}{1-e^t}} \times \sum_{\text{correct}} w_i^t \\ &= \sqrt{\frac{e^t}{1-e^t}} \times (1-e^t) = \sqrt{e^t(1-e^t)} \end{aligned}$$

$$\begin{aligned} \sum_{\text{wrong}} w_i^{t+1} &= \sqrt{\frac{1-e^t}{e^t}} \times \sum_{\text{wrong}} w_i^t \\ &= \sqrt{\frac{1-e^t}{e^t}} \times (e^t) = \sqrt{e^t(1-e^t)} \end{aligned}$$

Z = Normalization factor

$$w_i^{t+1} = \frac{w_i^t}{Z} \times \begin{cases} \sqrt{\frac{e^t}{1-e^t}} & \text{correct} \\ \sqrt{\frac{1-e^t}{e^t}} & \text{wrong} \end{cases}$$

$$Z = \sum_{\text{wrong}} w_i^{t+1} + \sum_{\text{correct}} w_i^{t+1} = 2\sqrt{e^t(1-e^t)}$$

$$w_i^{t+1} = \frac{w_i^t}{2} \times \begin{cases} \frac{1}{1-e^t} & \text{correct} \\ \frac{1}{e^t} & \text{wrong} \end{cases}$$

Sum Again

$$\sum_{correct} w_i^{t+1} = \frac{1}{2} \frac{1}{1 - e^t} \sum_{correct} w_i^t = \frac{1}{2}$$

$$\sum_{wrong} w_i^{t+1} = \frac{1}{2} \frac{1}{e^t} \sum_{wrong} w_i^t = \frac{1}{2}$$

Example

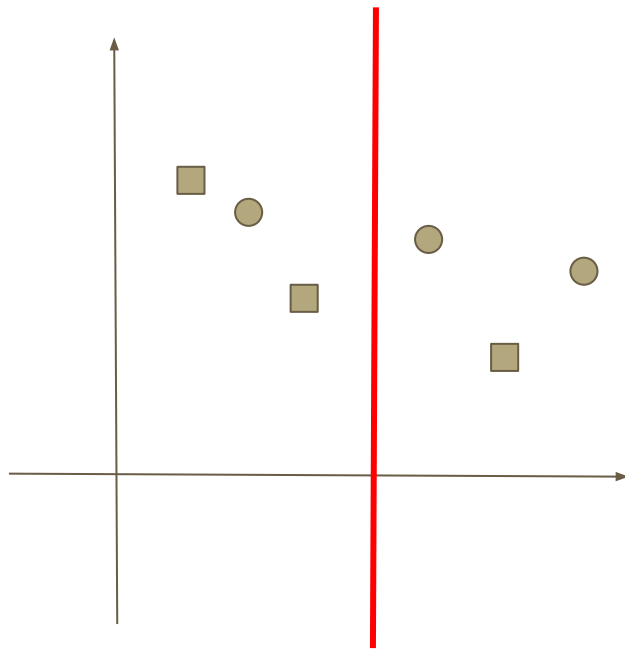
Two instances are wrongly classified

Error = ?

Total weight = ?

Accuracy = ?

What will be the weights after update?



distribution  $\sum_{all} w_i = 1$

We will always try to keep it as a distribution